

AWS Cloud Practitioner Essentials

Module 1: Introduction to Amazon Web Services – Notes

1 The Client–Server Model

- **Client:** makes a request (the customer)
- **Server:** receives the request, validates it, and returns a response (in AWS, typically a *virtual server*)
- This client–server pattern underlies most cloud computing interactions.

1.1 Pay-as-you-go pricing

- Traditional on-premises: pay upfront for fixed capacity, whether used or not.
- AWS: pay only for resources actually consumed.
- Provision resources instantly when needed; deprovision instantly to stop billing.

2 What Is Cloud Computing?

Cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing.

- **On-demand:** use resources as needed; stop paying immediately when deprovisioned.
- **Delivery:** resources live in data centers (buildings housing servers, with redundant power/cooling/security). With AWS, you use data centers you don't own or manage.
- **Over the internet:** access and manage infrastructure remotely via a browser, from anywhere.
- **Pay-as-you-go:** no contracts, no upfront investment, deprovision anytime.

2.1 Brief history

- Early 2000s: Amazon.com built internal tools to scale its e-commerce infrastructure.
- 2003: Amazon realizes this capability could be valuable to other businesses.
- Nov 2004: AWS launches its first public service – Amazon Simple Queue Service (SQS).
- 2006: Amazon S3 (storage) and Amazon EC2 (compute) launch.
- Today: AWS powers a large portion of the internet, serving millions of customers globally.

3 Cloud Deployment Types

Type	Where it runs	Why choose it
Cloud	Fully in AWS / cloud	Flexibility, scalability, no hardware to manage
On-premises	Fully in your own data center	Dedicated resources, low latency, compliance comfort
Hybrid	Both, working together	Legacy or regulatory needs combined with cloud benefits

Note: Multi-cloud deployments (using more than one cloud provider) are also considered a form of hybrid deployment.

Example scenario: A charity keeps sensitive data on-premises for compliance, but uses cloud resources to scale quickly for seasonal demand spikes → this is a **hybrid deployment**.

4 Six Benefits of the AWS Cloud

1. **Trade fixed expenses for variable expenses** – no large upfront investment; pay only for what you use.
2. **Benefit from massive economies of scale** – AWS buys hardware in bulk and passes savings to customers.
3. **Stop guessing capacity** – scale up or down in minutes instead of over- or under-provisioning hardware.
4. **Increase speed and agility** – spin up test environments quickly; delete them (and stop paying) if an idea doesn't work.
5. **Stop spending money running and maintaining data centers** – no racking, stacking, or powering servers yourself.
6. **Go global in minutes** – deploy to AWS Regions worldwide instead of building your own data centers abroad.

5 AWS Global Infrastructure

5.1 Key definitions

- **High availability:** application stays accessible with minimal downtime; if one component fails, another takes over.
- **Fault tolerance:** system continues operating even if *multiple* components fail; resilience built into every layer.

5.2 Infrastructure hierarchy

Region → Availability Zone (AZ) → Data Center

- **Region:** a geographic area (e.g., Paris, Tokyo, São Paulo, Dublin, Ohio), chosen to be close to customers.
- **Availability Zone (AZ):** an isolated location within a Region. Each Region has 3 or more AZs, physically separated so a disaster cannot take out more than one.
- **Data Center:** one or more physical buildings within each AZ, with redundant power, networking, and connectivity.

Note: For full disaster resilience, businesses often operate across *multiple Regions*, allowing failover if an entire Region has issues.

6 The AWS Shared Responsibility Model

AWS = security <i>of</i> the cloud Customer = security <i>in</i> the cloud	
AWS Responsibilities	Customer Responsibilities
Physical layer (data center security, hardware access control)	Operating system (including patching)
Network layer	Applications
Hypervisor / virtualization layer (workload isolation)	Data (access control, encryption)

Important nuance: the exact split of responsibility can shift depending on the AWS service used – e.g., with EC2 you manage more of the stack (OS, patching); with fully managed services, AWS takes on more.

7 Case Study: E-Commerce Company Going Global

7.1 Using Global Infrastructure

- Distance from infrastructure to customers increases latency.
- Solution: deploy in Regions close to customer bases, e.g., `eu-west-1` (Ireland) for Europe and `ap-southeast-1` (Singapore) for Asia.
- Region naming pattern: `[continent code]-[direction]-[number]`.
- Benefit for startups: global reach without large upfront capital investment.

7.2 Applying high availability

- Deploy the same configuration across at least 2 AZs.
- If one AZ fails, traffic fails over to the other.

7.3 Applying the Shared Responsibility Model

- AWS secures the physical/network/hypervisor layers.
- The company focuses on securing its data, managing access, and meeting compliance requirements (e.g., credit card data regulations).